CHIRAG KULAR

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LinkedIn | GitHub | Portfolio | U.S. Permanent Resident

PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 8+ years of production experience building and supporting web applications, APIs, data platforms, and distributed workflows. Combines Python and TypeScript backend engineering with React/Next.js product development, PostgreSQL and MongoDB data modeling, and AWS event-driven systems. Experienced in system integration, performance testing, observability, production debugging, and translating complex workflows into reliable software. Works across architecture, implementation, automated testing, code review, and production support.

CORE TECHNICAL SKILLS

Full-Stack: Python, TypeScript, Node.js, React, Next.js, Angular, Java, REST APIs, application state management
Databases: PostgreSQL, MongoDB, SQL Server
Cloud & Observability: AWS Lambda, S3, API Gateway, Step Functions, SQS, EKS/Kubernetes, EC2, Glue, CloudWatch, Docker, Terraform, Datadog, Splunk
Tools & Frameworks: Lit, Storybook, Playwright, JUnit, GitHub Copilot, Claude, Codex, RAG, OpenAI APIs, AWS Comprehend Medical
Healthcare & Interoperability: EHR/EMR, FHIR R4, US Core, HL7v2, Firely Server, Rhapsody Integration Engine, clinical workflows

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Full Stack | ICF — Feb 2026 - Present

- Own full-stack modernization of an NCI biospecimen research platform, migrating legacy capabilities to Next.js, TypeScript, PostgreSQL, and AWS EC2 while improving page performance by 40%.
- Translate researcher and operational workflows into accessible, data-driven product features, partnering with product, design, engineering, and domain stakeholders from requirements through production.
- Use Datadog APM and distributed tracing to diagnose service behavior and improve production reliability and performance visibility.

Software Engineer - Contract | M&T Bank (TEKsystems) — Oct 2024 - Feb 2026

- Designed a modular TypeScript/Lit component platform for the Currency design system, with 30+ reusable components adopted across 10+ product teams, establishing consistent APIs, accessibility patterns, and Storybook documentation.
- Built automated scaffolding and Playwright coverage for shared components, accelerating delivery and reducing regression risk across consuming applications.
- Partnered with UX, design, and application teams to turn complex workflows into accessible interfaces; improved maintainability through reviews, standards, and documented patterns.

Senior Software Engineer - Backend | ICF — Jul 2023 - Oct 2024

- Built secure AWS microservices that ingest, validate, process, and export healthcare data for the CMS Innovation Center's EOM and GUIDE models using Node.js/Python, Lambda, S3, SQS, Step Functions, API Gateway, MongoDB, and EKS.
- Designed asynchronous REST APIs for FHIR submissions and status polling; integrated Firely Server validation and persisted workflow state and metadata for reliable, traceable processing.
- Implemented event-driven workflows for file intake, ClamAV malware scanning, eligibility and metadata validation, failure recovery, reprocessing, and result export.
- Performance-tested 150K FHIR resources and a 250K-record end-to-end workload; tuned Lambda concurrency, throttling, batching, and Step Functions distributed-map behavior.
- Evaluated EHRbase, FHIRPath rules, US Core R4 profiles, and provenance/history patterns to support interoperability and long-term data correctness.
- Created Splunk and CloudWatch dashboards for load balancers, Lambda, MongoDB, workflow duration, health, and errors; independently diagnosed pipeline and production issues.
- Owned services from technical design through operational readiness; reviewed code and provided architecture, testing, and maintainability guidance across the team.

Software Engineer | MatrixCare — Aug 2018 - Jun 2023

- Built and maintained clinician-facing workflows in MatrixCare's skilled-nursing EHR using Java/Spring MVC, TypeScript/Angular, REST APIs, SQL Server, and AWS.
- Delivered Medication Reconciliation and Orders enhancements in CareAssist that streamlined patient readmissions and improved workflow efficiency by 50%.
- Integrated LabSoft laboratory data with the SNF platform, added HL7 fields, and updated Rhapsody Integration Engine routes to improve clinical interoperability and data reliability.
- Designed and optimized T-SQL stored procedures with execution-plan analysis for clinical workflows, reporting, and SAP Crystal Reports.
- Built reusable Angular forms, services, directives, and pipes for complex EHR workflows; partnered with UX and clinical stakeholders to simplify end-user interactions.
- Applied AWS Comprehend Medical to extract medication frequency from pharmacy instructions and evaluated practical NLP use cases for clinical data.
- Led two interns on an Okta authorization proof of concept using React, Python, and AWS Lambda; led a micro-frontend evaluation for the legacy Java platform.
- Provided on-call production support, diagnosed urgent customer issues, refactored legacy code safely, and expanded JUnit coverage during feature delivery.

Software Engineer | iGATE (now Capgemini) **Nov 2013 - Jun 2015**

- Supported Oracle Applications R12 financial modules and PL/SQL-based data conversion, requirements analysis, and end-to-end quality assurance.

PROJECTS

- HL7-FHIR Mapping Agent (2026): Prototype for mapping synthetic HL7v2 and C-CDA samples into FHIR R4, with deterministic parsing, retrieval-assisted mapping, and bounded validation.
- Scan Tip (2025-2026): Built and published a SwiftUI app for tip calculations, receipt scanning, possible service-charge or included-gratuity alerts, saved dining history, and iCloud sync.

EDUCATION

M.S., Computer Science - Rochester Institute of Technology, 2018 | GPA: 3.46
B.E., Information Technology - University of Mumbai, 2013 | GPA: 3.4

CERTIFICATIONS

- AWS Certified Data Engineer - Associate
- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- Datadog Fundamentals
- Datadog APM & Distributed Tracing Fundamentals
- Microsoft Certified: Azure AI Fundamentals

PATENT

Systems and Methods for Providing Prompts on an Electronic Device - US-20230004865-A1, published Jan 2023 (patent pending)